

Arrays $\Rightarrow$ Collection of elements
having same datatype

int arr[5] =

0	1	2	3	4
		4		

arr[2] $\Rightarrow$ 4

Q Print all elements of an array.

func(arr, n) {

for (int i = 0 ; i < n ; i++)
 print(arr[i]);

}

Q

Given an array of size n .
Reverse the array.

$n = 5$
arr = [1 2 3 4 5]
 (i) (j)

 ⇓
[5 4 3 2 1]

int i = 0, j = n - 1
while (i < j) {

 int temp = arr[i];
 arr[i] = arr[j];
 arr[j] = temp;
 i++, j--;

}

No of iteration = $n/2$

T.C : $O(n)$

Q2 Given an array of size n . Rotate the array from right to left k times.

$$n = 5, k = 2$$

$$\text{arr} = [1 \ 2 \ 3 \ 4 \ 5]$$

$$k=1 \quad [5 \ 1 \ 2 \ 3 \ 4]$$

$$k=2 \quad [4 \ 5 \ 1 \ 2 \ 3]$$

for (int i = 1; i ≤ k; i++) {

int temp = arr[n-1];

for (int j = n-2; j ≥ 0; j--) {

arr[j+1] = arr[j];

}

arr[0] = temp;

}

$T_C = O(nk)$

$$arr = \begin{bmatrix} \overset{j}{1} & \overset{1}{\cancel{2}} & \overset{2}{\cancel{3}} & \overset{3}{\cancel{4}} & \overset{4}{\cancel{5}} \end{bmatrix} \quad temp = 5$$

$$k(n-1) \Rightarrow n/2 \neq k$$

$$T_C: O(n^2)$$

$$arr = [1 \ 2 \ 3 \ 4 \ 5]$$

$$\downarrow \text{reverse}(0, n-1) \quad k = 2$$

$$arr = [5 \ 4 \ 3 \ 2 \ 1]$$

$$\downarrow \text{reverse}(0, k-1)$$

$$arr = [4 \ 5 \ 3 \ 2 \ 1]$$

$$\downarrow \text{reverse}(k, n-1)$$

$$arr = [4 \ 5 \ 1 \ 2 \ 3]$$

```
void reverse (arr, int s, int e) {
```

```
    int i = s, j = e;  
    while (i < j) {
```

```
        int temp = arr[i];
```

```
        arr[i] = arr[j];
```

```
        arr[j] = temp;
```

```
        i++, j--;
```

```
    }
```

```
}
```

```
func rotate (arr[], n, k) {
```

```
    k = k % n;
```

```
    reverse (arr, 0, n-1); - n
```

```
    reverse (arr, 0, k-1); - k
```

```
    reverse (arr, k, n-1); - n-k
```

```
}
```

= ~~n + k~~ + ~~n - k~~

TC: O(n)

arr() = [1 2 3 4 5]

k	Sorted Array.
1, 6, 11	5 1 2 3 4
2, 7, 12	4 5 1 2 3
3, 8, 13	3 4 5 1 2
4, 9, 14	2 3 4 5 1
5, 10, 15	1 2 3 4 5

$$k = k \% n$$

Dynamic Arrays!

Dynamic Arrays in Different Languages

Java

```
ArrayList<String> al = new ArrayList<>(); //ArrayList is created
al.add("50"); //50 is inserted at the end
al.clear(); // al={}

for (int i = 0; i < al.size(); i++) {
    System.out.print(al.get(i) + " ");
}
//iterating the ArrayList
```

50 20

C++

```
vector<int> a; //vector is created

a.push_back(60);
//a = {10, 20, 30, 40, 50, 60} after insertion at end

a.clear(); // a={}

for(int i=0; i<a.size(); i++) {
    cout<<a[i];
}
//iterating the vector
```

Python

```
thislist = [] //created list

thislist.append("orange") //added orange at end

thislist.clear() //cleared the list

for i in range(len(thislist)):
    print(thislist[i]) //iterating on the list
```

10:35pm

$k=2$

1 2

3 4 5

↓ $rev(0, n-1)$

5 4 3 2 1

↓ $reverse(, n-1)$

3 4 5 1 2

$[x, n-1] \neq 1$